

Data Analytics Phase 1

1. Business or Industry Problem Identification

The Nature of the Problem

Hotels in the hospitality world deal with this ongoing issue where they can't really predict how customers will book rooms, especially when seasons change or depending on the type of hotel and who the guests are. It gets messy because of that, leading to a few big problems in running things day to day.

One thing is the seasonal demand, which is hard to pin down. Like, city hotels might be busy in summer for events, while resort one's peak in winter vacations, but a lot of places just stick to the same pricing and staff schedules without adjusting much. I think that causes overstaffing when it's slow, wasting money on labor, or not enough people when it's crazy busy, which upsets the guests.

Then there's the occupancy rates; forecasting those accurately is tricky because cancellations throw everything off. Say a hotel looks fully booked months ahead at 100 percent, but then 40 percent cancel, and suddenly it's way lower. Hotels don't always know which months fill up after all that drops out; it seems like they miss the real picture there.

Customer cancellations vary a lot too, based on how they book, if they paid a deposit, or stuff like their age or family situation. OTAs such as Expedia or Booking.com bring in tons of reservations but also a bunch that gets canceled later. People booking directly might stick to it more, and families with kids could cancel differently than couples or someone traveling alone. Right now, hotels lack good data to spot which groups are riskier or more reliable. Maybe that's something they need to figure out better.

These issues with seasons, forecasts messed up by cancellations, and those varying patterns all tie together. They end up cutting into revenue, ramping up costs from poor

planning, and making guests unhappy overall. It's not fully clear how to fix it all at once, but yeah.

Why the Problem Is Important

This problem matters for three main reasons:

1. **Revenue Impact:** If a hotel can just get one percent more people to stay or stop one percent of people from canceling, that can mean hundreds of thousands of dollars in revenue every year. Hotels are losing a lot of money around 15 to 20 percent of what they could be making because people cancel at the minute, and they do not do a good job of guessing how busy they will be during different times of the year.
2. **Operational Efficiency:** When hotels do not know what to expect during times of the year they might hire too many people when it is slow and waste money or they might not hire enough people when it is busy and that makes the guests wait a long time and the rooms are not clean. If hotels can make guests guess about how busy they will be, they can save money on labor and make the guests happier.
3. **Advantage:** Hotels that really understand their customers can do things that other hotels cannot recommend the right rooms to people and have special deals during certain times of the year. Hotels that use data to make decisions are the ones that do the best in a market with a lot of competition.
4. **Customer Experience:** When hotels know what people like, such as that families like Resort hotels and couples like to take weekend trips, they can make sure those people get the right rooms and the right amenities. This makes the guests happy. They are more likely to come back to the hotel again. Hotels that understand their customers, like Resort hotels and City hotels can make people happy. Get more repeat bookings.

Hotel Owners / Investors

Lose money from low occupancy rates, high cancellation volumes, and inefficient operational spending. Reduced profitability affects return on investment and long-term business sustainability.

Revenue Managers

Cannot set optimal room prices or design effective deposit policies without understanding seasonal demand patterns and cancellation behaviors by customer segment.

Marketing Teams

Waste advertising budgets targeting the wrong customer segments, the wrong seasons, or the wrong hotel types, resulting in low conversion rates and poor return on marketing spend.

Front Desk & Housekeeping Staff

Face understaffing during peak seasons, leading to overwork, long guest wait times, and lower service quality. During off-peak seasons, overstaffing leads to reduced work hours and job insecurity.

Customers

Experience poor service quality during understaffed peak periods, miss out on personalized room offers and amenities, and may encounter rigid cancellation policies that do not match their preferences.

Online Travel Agencies (OTAs)

May lose hotel partnerships if hotels reduce their reliance on OTA bookings due to high cancellation rates, potentially reducing OTA commission revenue.

Local Tourism Boards

Lack of data regarding peak demand periods and the origin of their visitors prevent them from effectively coordinating seasonal promotions, events or marketing campaigns with hotels.

How Data Analysis Can Help Generate Useful Insight or Support Better Decisions

Data analysis helps in the following specific ways:

- 1. Identifies peak and off-peak seasons** so hotels can raise prices during high demand and offer discounts during low demand.
- 2. Calculates true occupancy rates** after removing cancellations, so hotels know which months actually fill rooms.

- 3. Reveals which customer segments cancel most** so hotels can require deposits from high-risk booking channels.
- 4. Quantifies how deposit policies affect cancellations** so hotels can implement non-refundable options to reduce revenue loss.
- 5. Compares lead times between customers who cancel and those who stay** so hotels can adjust cancellation window policies.
- 6. Discovers what different customer types prefer** (families vs. couples, Resort vs. City hotels) so marketing and room allocation can be personalized.
- 7. Provides statistical evidence** (p-values, confidence intervals) so managers can make policy changes with confidence, not guesswork.
- 8. Reduces revenue loss** by matching pricing, staffing, and marketing to actual demand patterns.
- 9. Improves customer satisfaction** by tailoring room assignments and offers to segment-specific preferences.
- 10. Supports data-driven decision-making** across pricing, operations, marketing, and customer retention strategies.

2. Analytical Questions

Question 1 – Seasonal Trends

How do booking volumes vary by month and season across City hotels compared to Resort hotels?

This question demonstrates a classification problem with overlapping classes — how we can infer if the two hotel types have different demand patterns. Resort hotels, for example, could peak in summer; City hotels might remain steady all year. Knowing this allows hotels to plan their pricing and staffing.

Question 2 – Occupancy Rates

Which arrival month has the highest occupancy rate after removing canceled bookings?

Raw booking numbers can be misleading because many customers cancel before arrival. This question focuses on realized occupancy — the rooms that actually get used. Finding the true peak month allows hotels to maximize revenue during that period.

Question 3 – Cancellation Behavior

What is the average lead time for canceled bookings compared to non-canceled bookings?

Lead time means the number of days between when a customer books and when they arrive. I want to see if customers who cancel tend to book further in advance than those who actually show up. If yes, hotels could require deposits for early bookings.

Question 4 – Market Segments

Which market segment (Online TA, Offline TA, Direct, Corporate, or other) has the highest cancellation rate?

Not all booking channels are equal. Some might generate many reservations but also many cancellations. This question identifies the riskiest segments so hotels can adjust deposit policies or cancellation deadlines for those channels.

3. Data Source(s)

Dataset 1: Hotel Booking Dataset

Source: Kaggle – saadharoon27/hotel-booking-dataset

Link: <https://www.kaggle.com/datasets/saadharoon27/hotel-booking-dataset>

Size and Structure: 119,390 rows with 35 columns (119,390, 35). comprehensive booking information.

Key Variables:

1. hotel
2. is_canceled
3. lead_time
4. arrival_date_year
5. arrival_date_month
6. arrival_date_week_number
7. arrival_date_day_of_month
8. stays_in_weekend_nights
9. stays_in_week_nights
10. adults
11. children
12. babies
13. meal
14. country

15. market_segment 16. distribution_channel 17. is_repeated_guest
18. previous_cancellations 19. previous_bookings_not_canceled 20. reserved_room_type
21. assigned_room_type 22. booking_changes 23. deposit_type 24. agent 25. company
26. days_in_waiting_list 27. customer_type 28. adr 29. required_car_parking_spaces
30. total_of_special_requests 31. reservation_status 32. reservation_status_date
33. name 33. email 34. phone-number 35. credit_card

Why It Is Appropriate: This dataset appears to be a version of the classic Hotel Booking Demand dataset. It contains almost all columns needed for the six questions, including hotel type, cancellation, lead time, market segment, deposit type, ADR, stays, children, and special requests.

Anticipated Limitations:

- Missing values in children column (approx 4 rows)
- Missing values in country column (approx 488 rows)
- Missing values in agent column (approx 16,340 rows)
- Missing values in company column (112,593 rows — mostly empty)
- A lot of missing values

Key Variables :

- Booking ID (unique identifier)
- Number of adults (n_adults)
- Number of children (n_children)
- Weekend nights (weekend_nights)
- Week nights (week_nights)
- Meal plan (meal_plan)
- Car parking space required (car_parking_space)
- Room type reserved (room_type_reserved)
- Lead time (lead_time)
- Arrival year (arrival_year)

- Arrival month (arrival_month)
- Arrival date (arrival_date)
- Market segment type (market_segment_type)
- Repeated guest (repeated_guest)
- Previous cancellations (no_of_previous_cancellations)
- Previous bookings not canceled (no_of_previous_bookings_not_canceled)
- Average price per room (avg_price_per_room)
- Special requests (no_of_special_requests)
- Booking status (booking_status) — indicates if canceled or not

Why It Is Appropriate: This dataset is specifically designed for predicting customer cancellations. It contains all the key columns needed for this project: lead time, market segment, repeated guest, special requests, cancellation status, children, stays (weekend/week nights), and average price. The data has no missing values, which makes preprocessing easier .

Anticipated Limitations:

- No hotel type column (City vs. Resort) — all data appears to be from one hotel or hotel chain
- No deposit type column
- No distribution channel column
- Room types are encoded (Room_Type 1, 2, 3, etc.) with no description
- Limited to 19 columns (less detailed than Dataset 5)

4. Preliminary Analysis Plan

The Hotel Booking Dataset has about 119,000 rows and 32 columns.

1. Import Libraries and Load the Data

We are going libraries as Pandas to read data set and do a lot of analysis and handle missing data,

Import matplotlib for data visualization and clear see, also seaborn for same reason

And import numpy for the log (if needed)

scipy.stats as stats (for hypothesis testing)

2. Inspect the Data

will use .info(), .head(), and .describe() to understand the structure, data types, and basic statistics. I will check for missing values, duplicates, and incorrect data types.

3. Handle Missing Data

Based on my review, missing values exist in four columns:

- children column: 4 missing rows. I will fill these with 0 (assuming no children).
- country column: 488 missing rows. I will keep these as "Unknown" since country is not required for my main questions and drop it or even if needed I will just fill with mode.
- agent column: 16,340 missing rows. I will drop this column because it is not needed for my questions.
- company column: 112,593 missing rows. I will drop this column for the same reason

4. Convert Data Types

- hotel – Convert to category. This column only has two values (City Hotel and Resort Hotel), so category is the correct type.
- is_canceled – Keep as integer. 0 means not canceled, 1 means canceled. Integer works well for calculations.
- lead_time – Keep as integer. This is the number of days between booking and arrival.
- arrival_date_year – Keep as integer. Years like 2015 and 2016 are numeric values.
- arrival_date_month – Convert to category. Month names like January and February should be categories, not strings.

- arrival_date_week_number – Keep as integer. Week numbers from 1 to 53 are numeric.
- arrival_date_day_of_month – Keep as integer. Days from 1 to 31 are numeric.
- stays_in_weekend_nights – Keep as integer. Count of weekend nights is numeric.
- stays_in_week_nights – Keep as integer. Count of week nights is numeric.
- adults – Keep as integer. Number of adults is numeric.
- children – Convert to integer after filling missing values. This column currently has float type because of 4 missing rows. After filling those with 0, I will convert to integer.
- babies – Keep as integer. Number of babies is numeric.
- meal – Convert to category. Meal plan codes (BB, HB, FB, etc.) are categorical.
- country – Convert to category. Country codes like PRT and GBR are categorical.
- market_segment – Convert to category. Booking channels like Online TA and Direct are categorical.
- distribution_channel – Convert to category. How the booking was made is categorical.
- is_repeated_guest – Keep as integer. 0 for first-time guest, 1 for repeat guest.
- previous_cancellations – Keep as integer. Count of prior cancellations is numeric.

- previous_bookings_not_canceled – Keep as integer. Count of prior successful stays is numeric.
- reserved_room_type – Convert to category. Room type codes (A, B, C, etc.) are categorical.
- assigned_room_type – Convert to category. Room type actually assigned is categorical.
-
- booking_changes – Keep as integer. Number of changes made to the booking is numeric.
-
- deposit_type – Convert to category. Three values: No Deposit, Non Refundable, Refundable.
-
- agent – Drop this column. Too many missing values (16,340 rows) and not needed for my six questions.
-
- company – Drop this column. Too many missing values (112,593 rows) and not needed for my six questions.
-
- days_in_waiting_list – Keep as integer. Number of days the customer waited is numeric.
-
- customer_type – Convert to category. Values include Transient, Contract, Group, and Transient-Party.
-
- adr – Keep as float. Average daily rate is a decimal number representing price.
-
- required_car_parking_spaces – Keep as integer. 0 or number of spaces requested.
-
- total_of_special_requests – Keep as integer. Count of special requests made by the customer.
-
- reservation_status – Convert to category. Values include Check-Out, Canceled, and No-Show.

-
- reservation_status_date – Convert to datetime. This is a date column and should be treated as such for time-based filtering.
-
- name – Drop this column. This is personally identifiable information and not needed for analysis.
-
- email – Drop this column. This is personally identifiable information and not needed for analysis.
-
- phone-number – Drop this column. This is personally identifiable information and not needed for analysis.
- credit_card – Drop this column. This is personally identifiable information and not needed for analysis.

5. Remove Duplicates

- Check for and remove duplicate rows to avoid redundancy.

6. Filter Out Erroneous or Unwanted Records

- I will check for impossible or illogical values in the dataset. Below is each column I will inspect, the condition it must meet, and the action I will take if the condition is violated.
-
- adults – Must be at least 1. Every booking should have at least one adult. If adults is 0 or negative, I will drop that row.
-
- children – Must be 0 or greater. After filling missing values with 0, there should be no negative values. If any negative values exist, I will set them to 0.
-
- babies – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- lead_time – Must be 0 or greater. Lead time cannot be negative because it represents days between booking and arrival. If any negative values exist, I will drop that row.
-
- adr – Must be greater than 0. Average daily rate is a price and cannot be zero or negative. If adr is 0 or negative, I will drop that row.
-

- stays_in_weekend_nights – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- stays_in_week_nights – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- total_nights – This is a calculated column (stays_in_weekend_nights + stays_in_week_nights). It must be at least 1. A booking with zero total nights does not make sense. If total_nights is 0, I will drop that row.
-
- booking_changes – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- previous_cancellations – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- previous_bookings_not_canceled – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- days_in_waiting_list – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- required_car_parking_spaces – Must be 0 or greater. If any negative values exist, I will set them to 0.
-
- total_of_special_requests – Must be 0 or greater. If any negative values exist, I will set them to 0

7. Feature Engineering

- total_nights – weekend nights + week nights. This gives total length of stay.
-
- season – Map months to Winter, Spring, Summer, or Fall. This helps analyze seasonal trends.
-
- stay_category – Short (1-2 nights), Medium (3-6 nights), or Long (7+ nights). This compares cancellation patterns by stay length.

-
- lead_time_category – Very Early (180+ days), Early (30-179 days), or Last Minute (under 30 days). This shows if early bookers cancel more.
-
- family_booking – True if children or babies are more than 0. This identifies family bookings for preference analysis.
-
- realized_occupancy – True if booking was not canceled. This calculates true occupancy rates.
-
- month_num – Convert month names to numbers 1-12. This ensures months sort correctly on charts.
-
- is_weekend_only – True if weekend nights are more than 0 and week nights are 0.
-
- is_weekday_only – True if week nights are more than 0 and weekend nights are 0.

8. Outlier Detection (Optional)

Columns to check: lead_time, adr, total_nights, previous_cancellations

By box plot and calculate IQR and get the outliers

Exploratory data analysis.

Univariate Analysis (Single Variables)

I will examine one variable at a time to understand its distribution.

Booking volume by hotel type – I will count how many bookings each hotel type has (City Hotel vs. Resort Hotel). This shows which hotel type is more popular.

Cancellation rate overall – I will calculate the percentage of bookings that were canceled. This gives a baseline before breaking down by segments.

Distribution of lead_time – I will use a histogram to see if most bookings are made far in advance or last minute.

Distribution of adr – I will use a histogram to see the typical price range per room. I expect most values to cluster around a common price with some expensive outliers.

Distribution of total_nights – I will use a histogram to see if short stays or long stays are more common.

Distribution of deposit_type – I will count how many bookings use No Deposit, Non Refundable, or Refundable deposits.

Distribution of market_segment – I will count bookings by Online TA, Offline TA, Direct, Corporate, and other segments.

Distribution of family_booking – I will count how many bookings are family bookings (with children) vs. non-family bookings.

Bivariate Analysis (Two Variables)

I will examine relationships between two variables to answer my questions.

Q1: Booking volumes by month and hotel type – I will create a grouped bar chart showing monthly booking counts for City Hotels and Resort Hotels side by side. This reveals seasonal patterns.

Q2: Occupancy rate by month – I will calculate the percentage of non-canceled bookings for each month. The month with the highest percentage is the answer.

Q3: Lead time for canceled vs. non-canceled – I will use side-by-side box plots to compare lead time distributions between customers who canceled and those who stayed.

Q4: Cancellation rate by market segment – I will calculate the cancellation percentage for each market segment and sort from highest to lowest. This identifies the riskiest segments.

Q5: Cancellation rate by deposit type – I will calculate the cancellation percentage for No Deposit, Non Refundable, and Refundable bookings. This shows how deposit policies affect cancellations.

Q6: Family preference for hotel type – I will calculate what percentage of family bookings go to Resort hotels vs. City hotels. I will do the same for non-family bookings. Comparing these percentages answers whether families prefer Resorts.

Lead time by market segment – I will calculate average lead time for each market segment to see if certain segments book earlier than others.

ADR by hotel type – I will compare average daily rate between City and Resort hotels to see which is more expensive.

Multivariate Analysis (Three or More Variables)

I will examine complex relationships involving multiple variables.

Heatmap of cancellation rates by month and hotel type – I will create a heatmap with months on one axis, hotel type on the other, and cancellation rate as the color. This shows which months are risky for each hotel type.

Correlation matrix – I will create a correlation matrix for numeric variables including lead_time, adr, total_nights, total_of_special_requests, previous_cancellations, and is_canceled. This reveals which variables move together.

Pivot table of average lead time by market segment and cancellation status – This shows whether canceled bookings in each segment have different lead times than non-canceled bookings.

Cancellation trend over time – I will group bookings by arrival year and month, then calculate cancellation rate for each month. A line chart will show whether cancellations are increasing, decreasing, or staying stable.

Family bookings across seasons – I will calculate the percentage of family bookings in each season. This shows whether families prefer traveling during specific times of the year.

Data visualization.

Visualization for Q1: Booking volumes by month and hotel type

Chart type: Grouped bar chart or line chart

What it shows: Monthly booking counts for City Hotels and Resort Hotels side by side

Why this chart: A grouped bar chart makes it easy to compare patterns between the two hotel types across all 12 months

X-axis: Month (January to December in correct order)

Y-axis: Number of bookings

Colors: One color for City Hotel, another for Resort Hotel

Visualization for Q2: Occupancy rate by month

Chart type: Bar chart

What it shows: Realized occupancy percentage for each month after removing cancellations

Why this chart: A simple bar chart clearly shows which month has the tallest bar = highest occupancy

X-axis: Month (January to December)

Y-axis: Occupancy rate (0% to 100%)

Highlight: The highest month will stand out visually

Visualization for Q3: Lead time for canceled vs. non-canceled

Chart type: Side-by-side box plots or violin plots

What it shows: Distribution of lead time for canceled bookings compared to non-canceled bookings

Why this chart: Box plots show the median, quartiles, and outliers, making it easy to compare distributions

X-axis: Cancellation status (Canceled, Not Canceled)

Y-axis: Lead time in days

Key insight: If the Canceled box is higher than the Not Canceled box, cancellers book earlier

Visualization for Q4: Cancellation rate by market segment

Chart type: Horizontal bar chart sorted from highest to lowest

What it shows: Cancellation percentage for each market segment (Online TA, Offline TA, Direct, Corporate, etc.)

Why this chart: Horizontal bars with labels are easy to read when segment names are long. Sorting from highest to lowest immediately shows the riskiest segment at the top

X-axis: Cancellation rate (0% to 100%)

Y-axis: Market segment name

Key insight: The top bar is the segment with the highest cancellation rate

Hypothesis testing

Hypothesis Test 1: Market Segment Cancellation Rates (from Q4)

Business question: Do customers who book through Online Travel Agencies cancel more frequently than those who book directly?

Null hypothesis (H_0): There is no difference in cancellation rate between the Online TA market segment and the Direct market segment.

Alternative hypothesis (H_1): The Online TA market segment has a significantly higher cancellation rate than the Direct market segment.

Why I chose this test: I am comparing proportions between two independent groups. The two-proportion Z-test is designed for exactly this situation. Both variables are categorical (market segment and cancellation status).

Statistical test: Two-proportion Z-test

What I will calculate:

- $\hat{p}_1$ = cancellation rate for Online TA segment
- $\hat{p}_2$ = cancellation rate for Direct segment
- Z-statistic
- P-value
- 95% confidence interval for the difference in proportions

Assumptions I will check:

- The two groups are independent (Online TA bookings are separate from Direct bookings)
- Each group has at least 10 successes (cancellations) and 10 failures (non-cancellations)

Significance level: $\alpha = 0.05$

Hypothesis Test 2: Deposit Type and Cancellation (from Q5)

Business question: Does requiring a Non Refundable deposit reduce cancellation rates compared to No Deposit?

Null hypothesis (H_0): There is no association between deposit type and cancellation status. The cancellation rate is the same across all deposit types.

Alternative hypothesis (H_1): There is a significant association between deposit type and cancellation status. Specifically, Non Refundable deposits have lower cancellation rates than No Deposit bookings.

Why I chose this test: Both variables are categorical. Deposit type has three categories (No Deposit, Non Refundable, Refundable). Cancellation status has two categories (canceled, not canceled). The chi-square test of independence determines whether the two categorical variables are related.

Statistical test: Chi-square test of independence

$\alpha = 0.05$

May use powerBI to make tables and visualization for user use.

